TEXORA · LECTURE SHEET

Selection of Yarn Count for Fabric of Varying GSM

A buyer specifies GSM; the mill must answer with a yarn count. This sheet derives the selection rule from the structure relations themselves — for knitted fabric, GSM at full relaxation collapses to a product of width density and tex — then applies the same thinking to woven cloth, where the count is chosen against ends, picks and crimp.

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1. The knitted derivation — stitch length cancels

Start from the two relations established in TE-3205 and used in Sheet 8: the GSM identity, and Munden's equilibrium for cotton jersey, CPC × l ≈ 56 (CPC per cm, l in mm) in the fully relaxed state.

$$GSM = WPC \times CPC \times l \times tex \div 100, \quad \text{with } CPC = 56 \div l$$

$$\Rightarrow GSM = WPC \times (56 \div l) \times l \times tex \div 100 = \mathbf{0.56 \times WPC \times tex}$$

The stitch length cancels. In the fully relaxed fabric, GSM is fixed by just two choices: **how many wales per cm the gauge delivers (WPC)** and **how heavy the yarn is (tex)**. Stitch length still matters — it sets CPC, tightness and hand — but it cannot rescue a wrong count: if the product 0.56 × WPC × tex is not the target GSM, no cam setting will make the relaxed fabric weigh what the order says. (Check against the course's own cloth: TE-3205's jersey, WPC 12, 19.7 tex → 0.56 × 12 × 19.68 = 132 g/m² — its measured GSM.)

WORKED EXAMPLE — CHOOSING A COUNT FOR 180 G/M²

Target: cotton single jersey, 180 GSM, on a machine whose relaxed WPC will be 13 per cm.
Required tex = 180 ÷ (0.56 × 13) = **24.7 tex** → choose **Ne 24 (24.6 tex)**.
Resulting GSM = 0.56 × 13 × 24.6 = **179 g/m²** — within tolerance of the target.
Now the stitch length is chosen for tightness, not weight: at K = 1.45, l = √24.6 ÷ 1.45 = **3.42 mm**, giving CPC = 56 ÷ 3.42 = **16.4 per cm**. Count first (for GSM), stitch length second (for structure) — the order of decisions is the lesson. Sheet 12 builds a complete production plan on exactly these numbers.

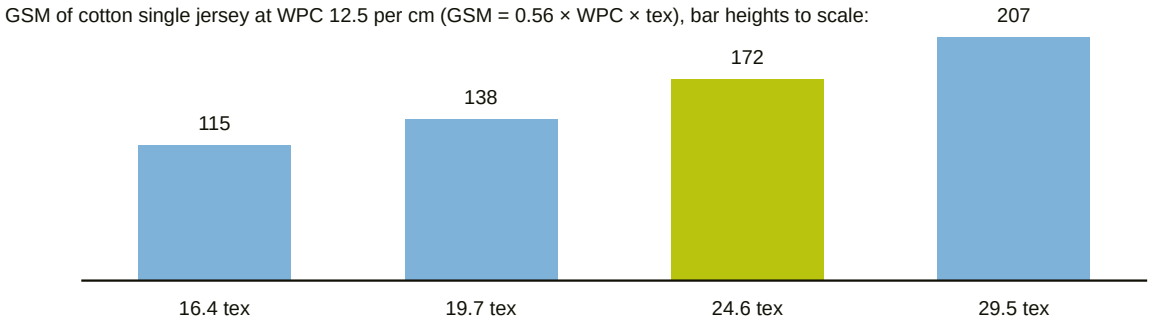


Figure 1 — GSM is a straight line in tex at fixed WPC: count selection is reading the target off this line.

2. The selection table and its limits

TARGET GSM (JERSEY)	MACHINE / WPC	TEX REQUIRED	COUNT CHOSEN	GSM OBTAINED
132	24 G / 12.0	19.6	Ne 30 (19.68)	132
180	28 G / 13.0	24.7	Ne 24 (24.6)	179
190	24 G / 11.5	29.5	Ne 20 (29.53)	190
220	20 G / 11.0	35.7	Ne 16 (36.9)	227 — re-trim WPC or accept

The last row teaches the boundary condition: counts come in the steps the spinning mill offers, and gauges fix WPC within a narrow band, so the product sometimes lands off-target. The planner's escapes, in order of preference: adjust WPC slightly by gauge or relaxation/finishing route; blend the difference in finishing (raising, compacting changes area, hence GSM); only as a last resort, accept a GSM at the edge of the buyer's tolerance — with the deviation calculated and declared, never discovered.

3. The woven parallel

For woven cloth the count choice runs through Sheet 5's formula with the densities fixed by the fabric design: GSM = 39.37 × tex × [EPI(1 + $crimp_1$) + PPI(1 + $crimp_2$)] ÷ 1,000 for equal warp/weft counts. For the Sheet 5 construction (EPI 100, PPI 80, crimps 8%/5%), the bracket is 100 × 1.08 + 80 × 1.05 = 192, so GSM = 39.37 × 192 × tex ÷ 1,000 = 7.56 × tex: Ne 30 (19.68) gives 149 ✓. To raise this cloth to 170 GSM at the same construction, tex must rise to 170 ÷ 7.56 = 22.5 → **Ne 26 (22.71 tex)**, giving 172 g/m². Same doctrine as knitting, one extra variable: densities and crimps share the stage with count,

so a woven GSM change can also be bought with picks — a count change is not the only lever, merely the strongest.

Glossary

Count selection — choosing yarn tex/Ne to hit a specified GSM at a given construction. **Munden relation** — CPC × l ≈ constant for a relaxed knitted structure family. **Tightness factor K** — √tex ÷ l; chosen after the count, to set structure. **Trim** — adjusting WPC/finishing to land a GSM the count steps cannot hit exactly.

Key points

• Knitted: GSM = 0.56 × WPC × tex at full relaxation — stitch length cancels; count is the GSM decision, l is the structure decision.
• Worked: 180 GSM at WPC 13 → 24.7 tex → Ne 24, l = 3.42 mm at K 1.45.
• Counts come in steps: land as close as the steps allow, then trim by WPC/finishing — declared, not discovered.
• Woven: GSM scales with tex at fixed construction (7.56 × tex for the reference cloth), but picks and crimp are levers too.

Practice

1. Select a count for 160 GSM cotton jersey at WPC 12 per cm, and give the GSM the chosen count actually delivers.
2. A mill knits the 180 GSM design of Section 1 but with Ne 20 (29.53 tex) by mistake, at the same WPC. What GSM comes off the machine, and can a stitch-length change fix it?
3. For the woven reference cloth, what tex gives 200 GSM at unchanged construction, and what standard count is nearest?

Answer key

1. tex needed = 160 ÷ (0.56 × 12) = **23.8 tex → Ne 24 (24.6 tex)** delivers 0.56 × 12 × 24.6 = **165 g/m²** (Ne 26, 22.71 tex, would give 153 — Ne 24 is the closer step).
2. GSM = 0.56 × 13 × 29.53 = **215 g/m²**. No: at full relaxation stitch length cancels from GSM — changing l changes CPC and tightness only. The fabric must be re-knitted in the right count (or the WPC changed by a different gauge); the error is in the yarn store, not the cam box.
3. tex = 200 ÷ 7.56 = **26.5 tex**; nearest standard count **Ne 22 (26.8 tex)**, giving 7.56 × 26.8 ≈ 203 g/m².